

VAIBHAV SHARMA

Bangalore, India

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SUMMARY

Backend and systems engineering student focused on building secure, scalable, and maintainable software systems. Experienced in designing backend architectures, transactional APIs, and event-driven workflows using Spring Boot, PostgreSQL, and asynchronous processing patterns.

EDUCATION

Mangalore Institute of Technology and Engineering
B.E. Computer Science and Engineering (CGPA: 9.43)

Sep 2023 – Jun 2027
Karnataka, India

TECHNICAL SKILLS

Languages: Java, Python, SQL, C
Backend: Spring Boot, Spring Security, Spring Data JPA, REST APIs, FastAPI, Flask, Express.js
Databases: PostgreSQL, MySQL, Redis, MongoDB, Firestore
Systems & DevOps: Linux, Docker, Git, GitHub Actions, Bash, Flyway, GCP
Concepts: RBAC, Distributed Systems Basics, Event-Driven Architecture, Rate Limiting, API Design, Database Modeling

INTERNSHIP

AI/ML Intern Sep 2025 – Nov 2025
Infosys Springboard Virtual

- Built backend APIs coordinating multiple ML inference models using Flask within an AI music generation system.
- Implemented model preloading and fallback logic, reducing initialization latency by **45%** and ensuring seamless inference workflows for high-concurrency requests.
- Designed REST endpoints for music generation, metadata storage, and genre classification backed by MongoDB.

Flutter Developer Intern Jul 2025 – Sep 2025
ClubChat Startup Remote

- Contributed to a production mobile application serving **8,000+** users.
- Developed event feed and real-time chat features using Flutter and Dart.
- Resolved **18+ stability issues** across two production releases through debugging and root-cause analysis.

PROJECTS

Appraisal Management System | Spring Boot, PostgreSQL, Hibernate 2026

- Designed and implemented a modular monolith backend for a college appraisal workflow management system.
- Built JWT-based authentication and domain-specific authorization workflows for faculty, reviewers, and administrators.
- Designed normalized PostgreSQL schemas, entity mappings, validation layers, and transactional APIs using Spring Boot and Hibernate.
- Applied SOLID design principles, input sanitization, and role-based access control to improve maintainability and backend security.

Arachnode: Automated Job Discovery & Outreach Platform

| Python, FastAPI, Redis Streams, PostgreSQL, Docker 2026

- Built a self-hosted **event-driven microservice platform** for automated job discovery, recruiter contact enrichment, and AI-assisted cold email drafting.
- Designed **6 independent services** with FastAPI, Scrapy, Playwright, Redis Streams, and PostgreSQL.
- Implemented **Redis consumer-group pipelines** with job deduplication and atomic state tracking.
- Developed an API gateway, scheduler-driven workflows, and analytics dashboard, cutting manual outreach effort by **80%+**.

- Academic Timetable Scheduling System** | Spring Boot, PostgreSQL, Docker 2026
- Designed and implemented a scheduling software system for automated timetable generation across **8 semesters**.
 - Analyzed software requirements and implemented constraint-based allocation logic to prevent faculty, room, and section conflicts.
 - Designed **14 normalized JPA entities**, interface mappings, and Flyway schema migrations with transactional rollback safety.
 - Performed integration testing and containerized deployment using Docker with secure role-based access control.
- Offline AI Assistant Shell** | C, Linux Syscalls, Seccomp 2025
- Designed and developed a secure C-based Linux utility program integrating an offline LLM using `llama.cpp`.
 - Implemented system-level software isolation using Seccomp syscall filtering, Linux namespaces, and resource constraints.
 - Built validation checks, syscall auditing, and debugging workflows to ensure safe execution in sandboxed environments.
- VulnGraph: Attack Path Visualization Prototype** | FastAPI, React, NetworkX 2026
- Built a prototype for visualizing vulnerability attack paths across Python codebases using graph traversal techniques.
 - Implemented AST-based source-to-sink taint tracking and interactive attack-path visualization using NetworkX and React force graphs.
 - Integrated natural language query handling and patch simulation workflows for security analysis demonstrations.
 - Designed graph-based risk scoring and Git-based vulnerability tracing workflows within a 24-hour team hackathon environment.

LEADERSHIP & COMMUNITY

- President, Coders Club** Oct 2025 – Present
MITE Campus
- Mentor students in Git, Data Structures, and backend development.
 - Organized **Hack n' Seek**, a hybrid technical event with **700+ online participants** and **180+ on-site attendees**.
 - Active participant in **GDG, cloud, AI, and cybersecurity communities**, regularly attending developer events, architecture workshops, and hackathons to stay current with modern engineering practices.
- Open Source Contributor & Project Admin** Feb 2026 – Present
GSSoC / ELUSOC / Aperture
- Serve as Project Admin for **Arachnode** under GSSoC, and ELUSOC, mentoring contributors and reviewing backend architecture, API workflows, and system integration changes.
 - Contributed to the GitCanvas open-source codebase under the Aperture 3.0 with **2 pull requests merged** after maintainer review.
 - Participate in GSSoC open-source collaboration workflows including issue triaging, contributor coordination, code review discussions, and feature planning.